
James Kenneth A. Tamaca

Blk 8 Lot 29 Villa Sta. Monica, Brgy. Sta. Monica • San Pablo City, Laguna, 400 • james2413tamaca@gmail.com • 09454554939

Professional Summary

Electronics Engineering graduate with hands-on experience in embedded systems development, LoRaWAN IoT solutions, and full-stack web development. Proven ability to lead projects from concept to deployment, optimize power consumption in battery-operated devices, and deliver real-world solutions in partnership with government agencies. Passionate about using technology for environmental protection and sustainable development.

Education

Laguna State Polytechnic University - San Pablo City Campus

San Pablo, Laguna

Bachelor of Science in Electronics Engineering

2022 - 2026

Development of a LoRaWAN-Enabled Per-Tree Monitoring Device for Geolocation and Forest Disturbance Detection
Best Thesis in Electronics Engineering

Relevant Coursework: Circuit Analysis, Signal Processing, Electronic System Design, Communications Engineering, Embedded Systems

Experience

Free Lance

San Pablo City, Laguna

Free Lance Tutor

October 2021 – June 2022

- Tutored students from Grade 1 to Grade 12 in mathematics, science, and English subjects
- Adapted teaching strategies to accommodate diverse learning styles and ability levels
- Developed customized lesson plans to address individual student weaknesses and improve academic performance

Special Program for the Employment of Students (SPES)

San Pablo City, Laguna

Administrative Assistant (Intern)

June 2023 – June 2026

- Supported daily operations of the local Barangay office through documentation and data collection
- Maintained accurate records and organized filing systems for administrative efficiency
- Collaborated with team members to complete tasks within established deadlines

City Engineering Office – San Pablo

San Pablo City, Laguna

Electronics Engineering - Intern

February 2026 – April 2026

- Assisted in the inspection and maintenance of city infrastructure electrical systems
- Participated in site surveys and data collection for engineering projects
- Supported the preparation of technical reports and documentation
- Collaborated with licensed engineers on ongoing city development projects
- Gained hands-on experience in government engineering operations and project management

Leadership & Activities

Association of Future Electronics Engineers (AFECE)

San Pablo City, Laguna

President

June 2025 – June 2026

- Led the organization of electronics engineering students, overseeing all activities and initiatives
- Coordinated with faculty and administration to plan and execute academic and professional development events
- Facilitated seminars, workshops, and outreach programs for students
- Managed the organization's budget and resources
- Represented the student body in engineering-related meetings and events
- Mentored and guided fellow students in academic and professional growth

Certifications

Introduction to Data Science | Cisco Networking Academy | *June 2026*

- Explained the promises and challenges of data analytics
- Understood the role of data in AI and Machine Learning
- Explored career pathways in Data Analytics

Introduction to Modern AI | Cisco Networking Academy | *July 2026*

- Developed proficiency in prompt engineering and LLM utilization
- Applied multimodal AI techniques for image and text processing
- Demonstrated use cases: idea generation, summarization, rewording, and grading

Introduction to Cybersecurity | Cisco Networking Academy | *May 2026*

- Identified common cyber threats, attacks, and vulnerabilities
- Implemented strategies for online safety and organizational protection

Yeastar Certified Technician - P-Series Software & Appliance Editions | Yeastar | *December 2025*

- Installed, configured, and maintained Yeastar P-Series IP PBX systems
- Troubleshoot telephony systems and resolved technical issues

AWS Solution Architect - Associate | Alison | *April 2024*

- Designed and deployed scalable cloud infrastructure solutions
- Gained proficiency in AWS core services including EC2, S3, and VPC

Professional Development

ECE Thrive 2025 Webinar Series

- Industry 5.0
- Generative AI and Prompt Engineering in the Academe and Industrial Workplace
- Electronics Professionals leading to a Greener Future
- Leadership in Engineering: Cultivating Visionaries for a Sustainable World

Skills & Interests

Technical:

- Programming Languages: C (Proficient), C++ (Proficient), Arduino (Proficient), Python (Intermediate), JavaScript (Intermediate), HTML/CSS (Intermediate), SQL (Intermediate)
- IoT & Embedded: LoRaWAN, LPWAN, RAK4631, MPU-6050, NEO-M8N GPS, I2C, UART, SPI, GPIO, Power Management
- Web Development: Flask, PostgreSQL, Leaflet, Chart.js, REST APIs, Full-Stack Development
- Hardware: Sensor Integration, Signal Conditioning, PCB Assembly, Battery Management
- Data & AI: Data Analytics, AI Fundamentals, Machine Learning Concepts, Prompt Engineering
- Tools & Platforms: TTN Console, MQTT, Git, Arduino IDE, Linux/Bash, Google Colab
- Networking: LoRa, AS923 Band, OTAA, RSSI/SNR Analysis, Gateway Configuration
- Office & Other: Microsoft Office Suite (Word, Excel, PowerPoint), AutoCAD (Basic)

Language:

- Filipino (Native)
- English (Fluent)

Interests:

- Electronics and IoT Innovation
- Environmental Technology and Sustainability
- Teaching and Mentoring
- Continuous Learning and Professional Development